Exploratory Data Analysis

Presented by :

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Exploratory Data Analysis (EDA)

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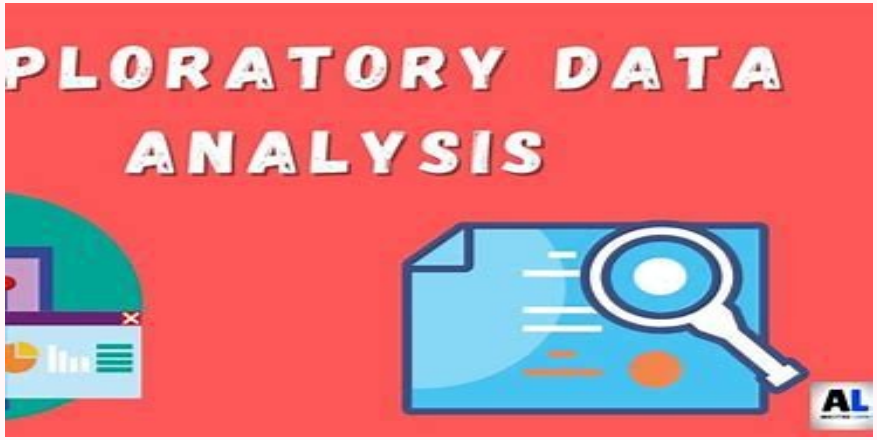
- A process of analyzing datasets to summarize their main characteristics using visualizations and statistical techniques.
- Helps in identifying patterns, trends, and anomalies.
- Why is it important?
- Ensures data quality.
- Helps in feature selection and model selection.
- Detects missing values and outliers.



Steps in EDA

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- **Understanding the Data**
- Checking dataset structure (columns, rows, types).
- Identifying categorical vs. numerical variables.
- **Handling Missing Values**
- Checking for missing values.
- Imputation strategies (mean, median, mode, or removal).
- **Descriptive Statistics**
- Summary statistics: mean, median, mode, min, max, standard deviation.
- **Data Visualization**
- Histograms, box plots, scatter plots, and correlation heatmaps.
- **Outlier Detection**

Understanding Data Types

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- **Categorical Data:** Gender, country, product categories.
- **Numerical Data:** Age, income, price.
- **Ordinal Data:** Customer ratings (low, medium, high).

Slide 5: Summary Statistics

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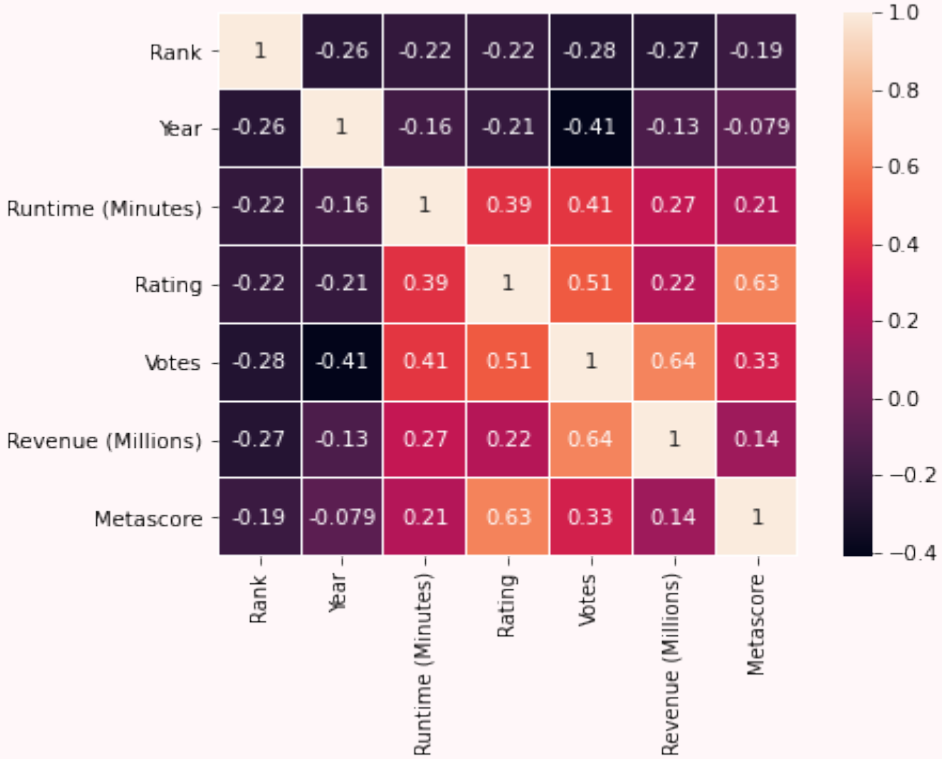
- **Mean, Median, Mode**

- **Variance & Standard Deviation**

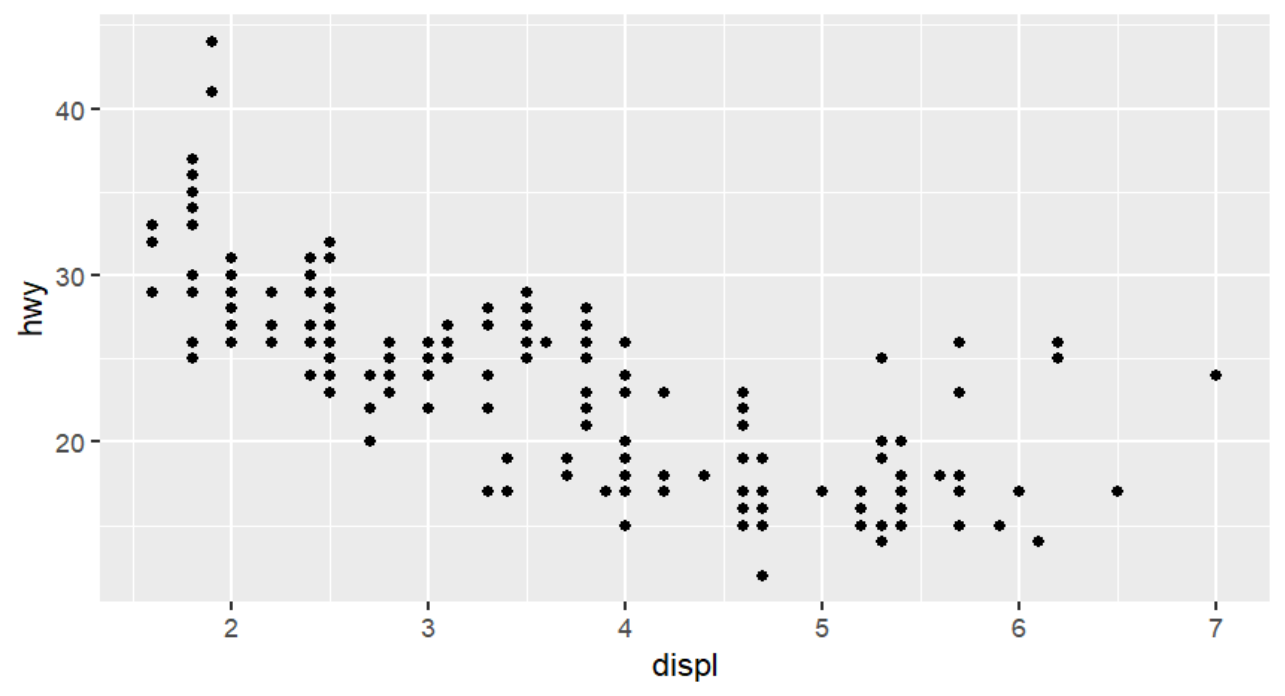
	Rank	Year	Runtime (Minutes)	Rating	Votes	Revenue (Millions)	Metascore
count	1000.000000	1000.000000	1000.000000	1000.000000	1.000000e+03	872.000000	936.000000
mean	500.500000	2012.783000	113.172000	6.723200	1.698083e+05	82.956376	58.985043
std	288.819436	3.205962	18.810908	0.945429	1.887626e+05	103.253540	17.194757
min	1.000000	2006.000000	66.000000	1.900000	6.100000e+01	0.000000	11.000000
25%	250.750000	2010.000000	100.000000	6.200000	3.630900e+04	13.270000	47.000000
50%	500.500000	2014.000000	111.000000	6.800000	1.107990e+05	47.985000	59.500000
75%	750.250000	2016.000000	123.000000	7.400000	2.399098e+05	113.715000	72.000000
max	1000.000000	2016.000000	191.000000	9.000000	1.791916e+06	936.630000	100.000000

Feature Correlation

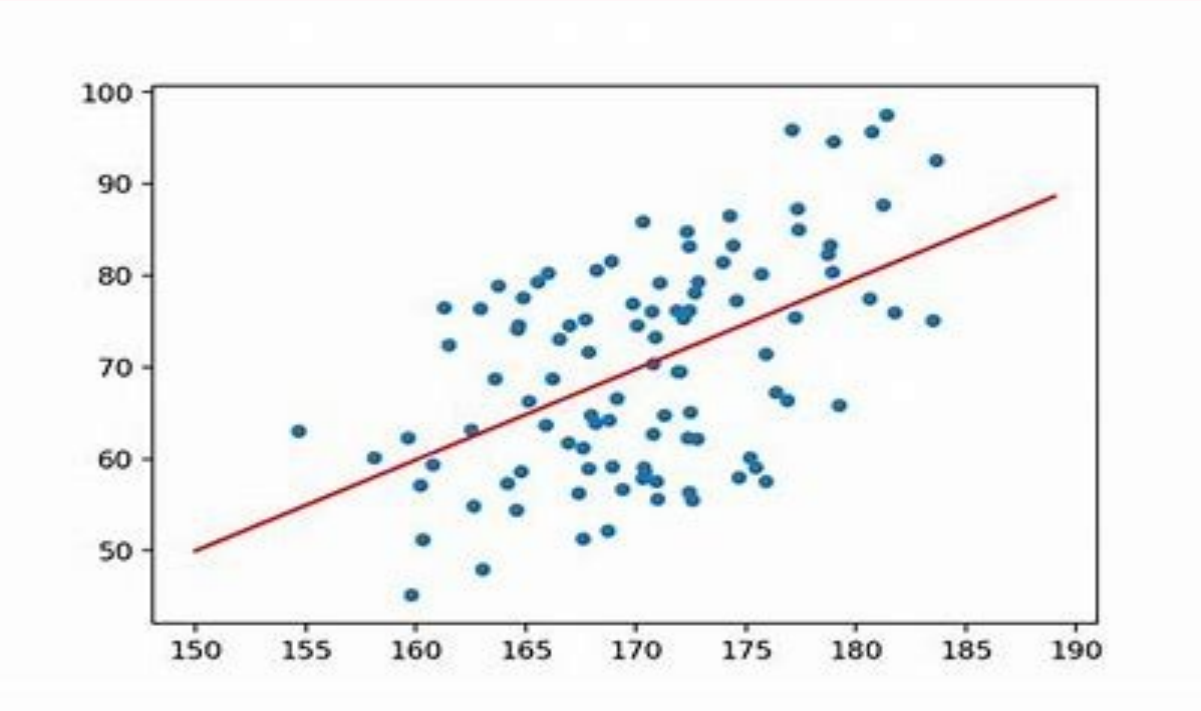
- **Feature Correlation**
- **Heatmaps** to show relationships between numerical variables.
- Helps identify which variables are highly correlated, allowing you to remove redundant features.



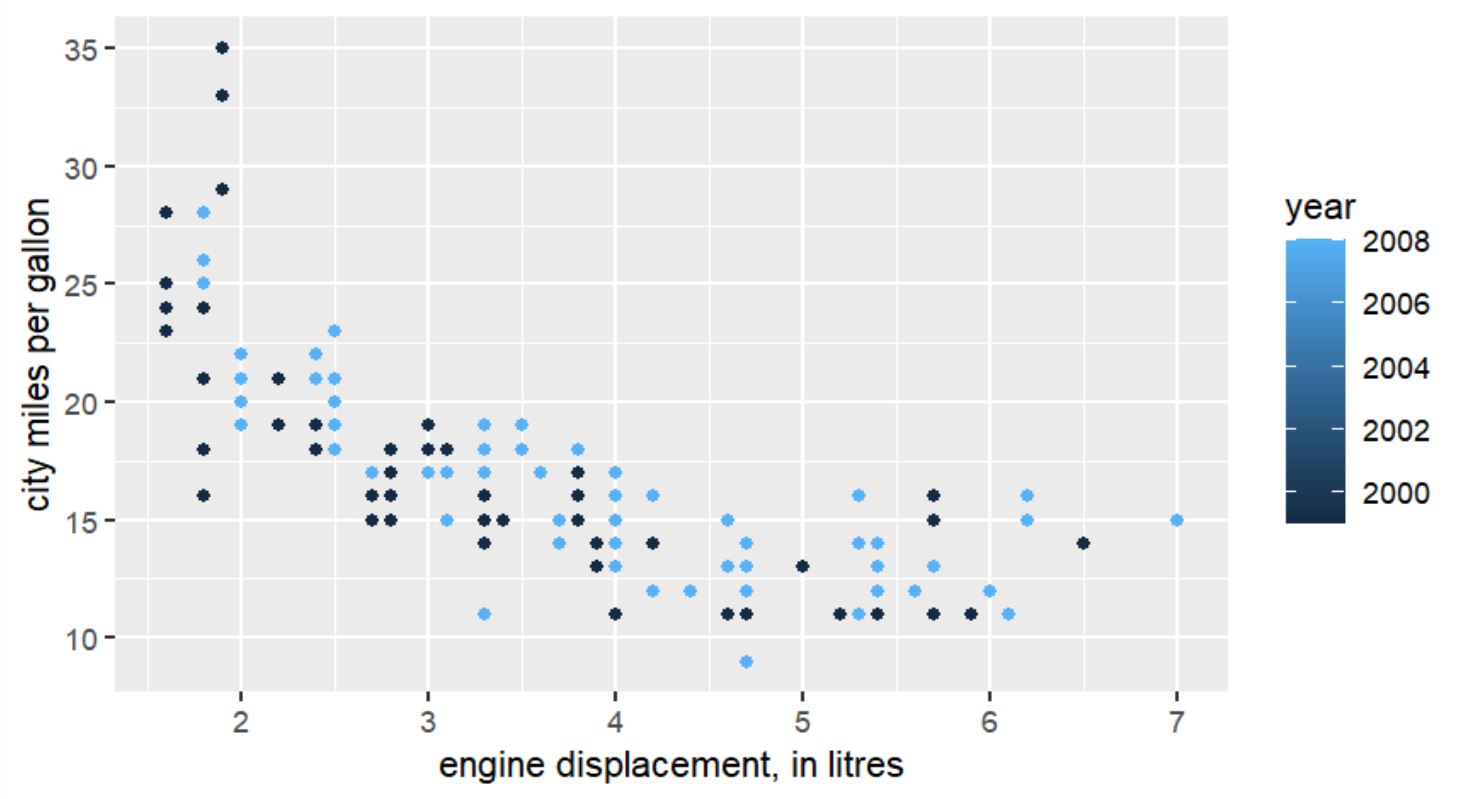
Scatter plot



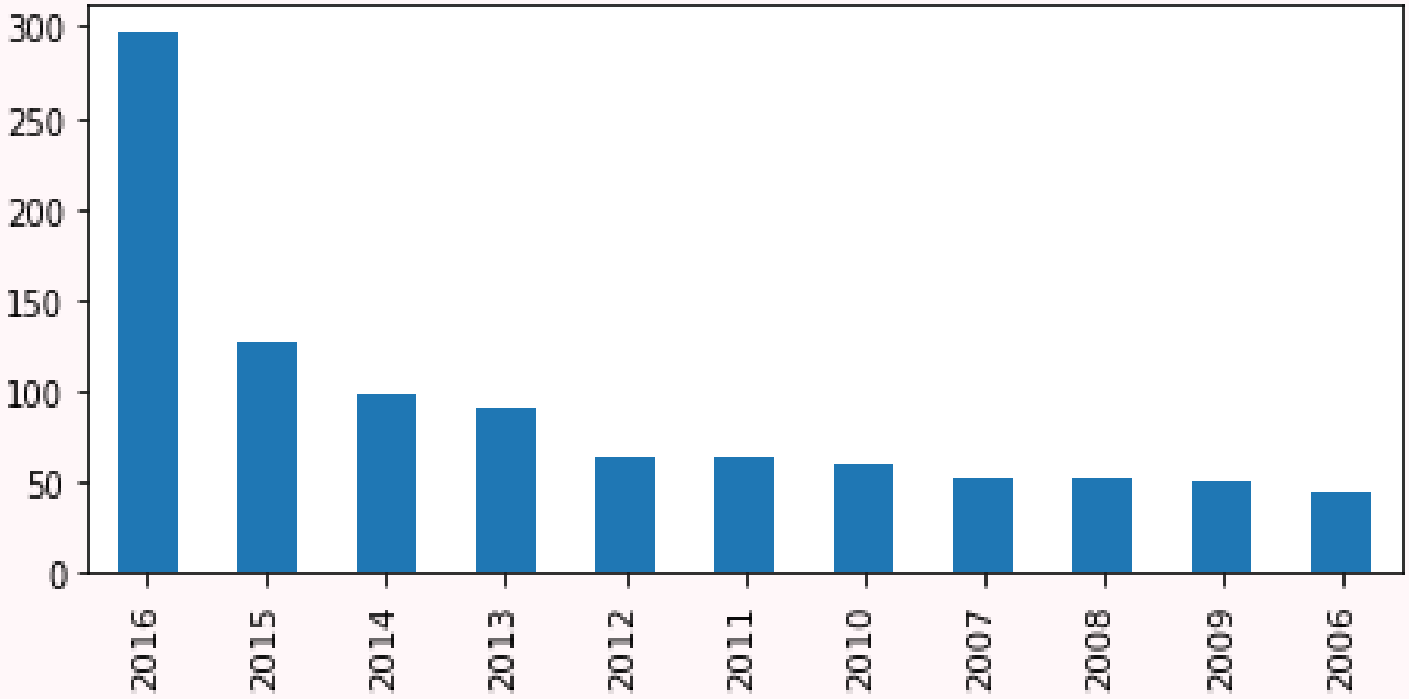
Scatter plot



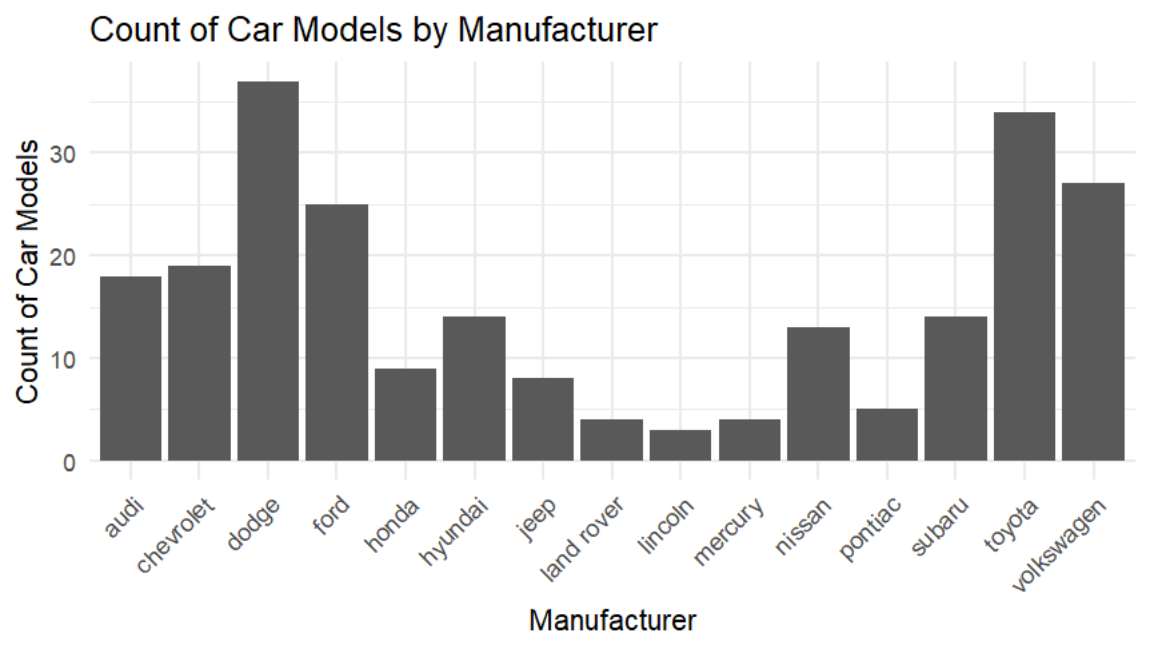
Scatter plot



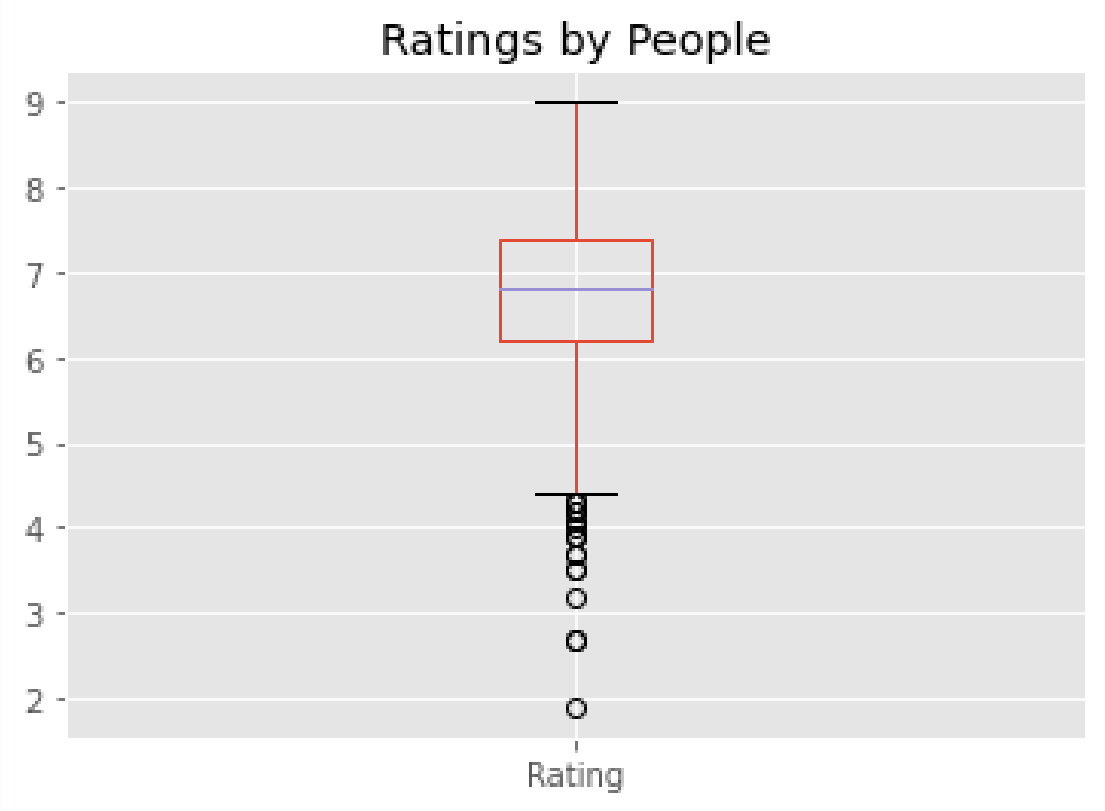
Bar plot


300
250
200
150
100
50
0
2016
2015
2014
2013
2012
2011
2010
2007
2008
2009
2006

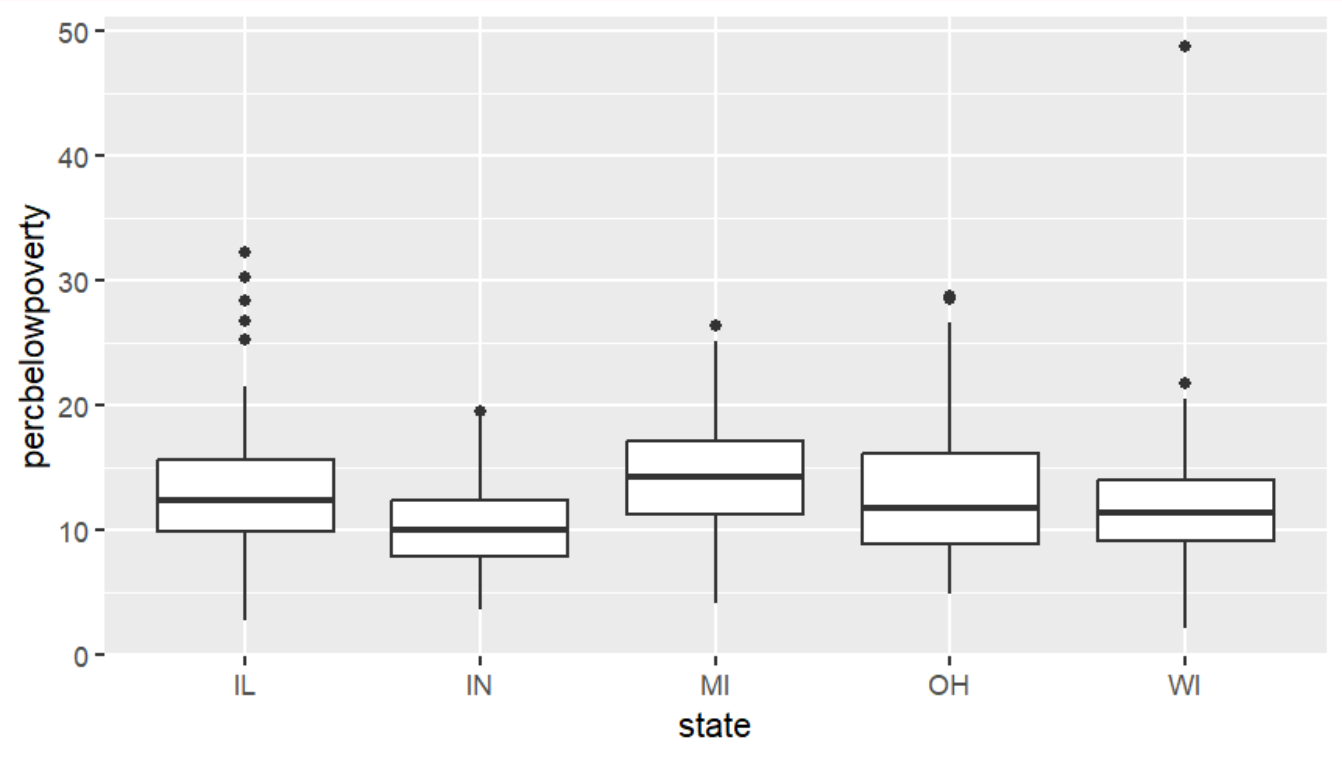
Bar plot


Count of Car Models by Manufacturer
Count of Car Models
0
10
20
30
audi
chevrolet
dodge
ford
honda
hyundai
jeep
land rover
lincoln
mercury
nissan
pontiac
subaru
toyota
volkswagen
Manufacturer

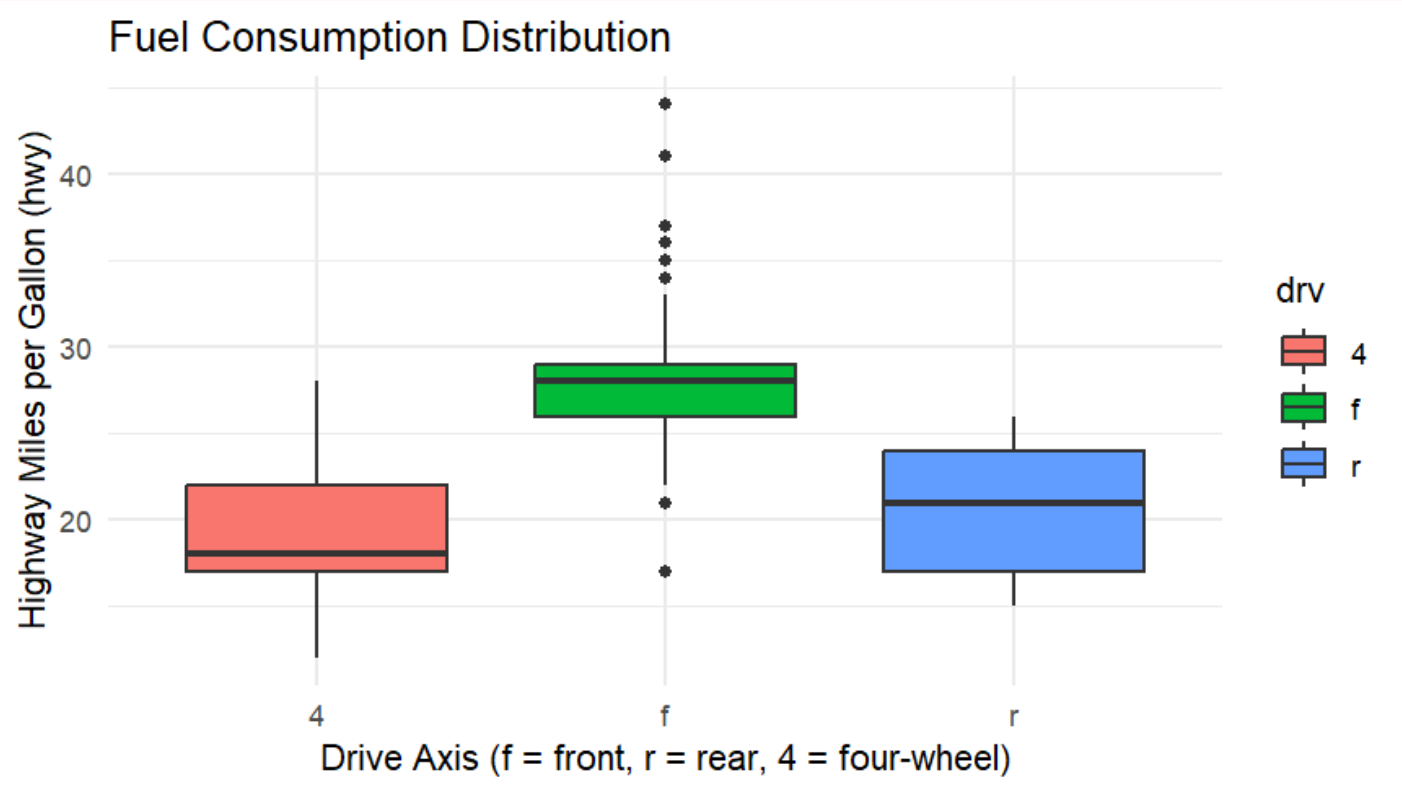
Box plot



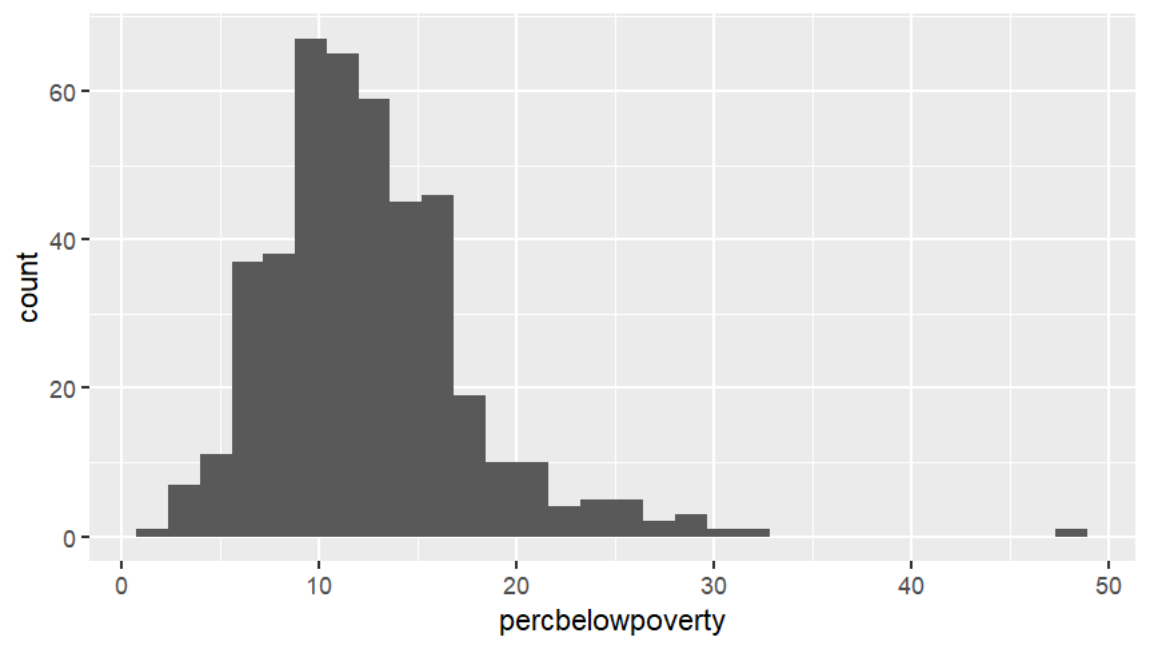
Box plot



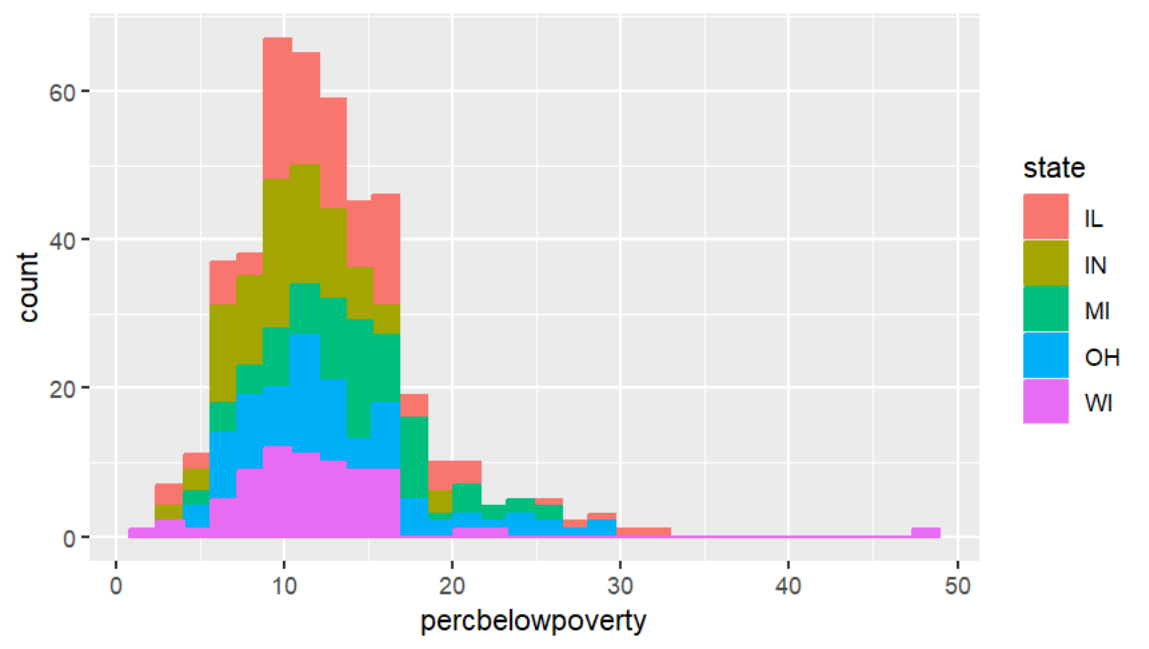
Box plot


Fuel Consumption Distribution
Highway Miles per Gallon (hwy)
40
30
20
4
f
r
Drive Axis (f = front, r = rear, 4 = four-wheel)
drv
4
f
r

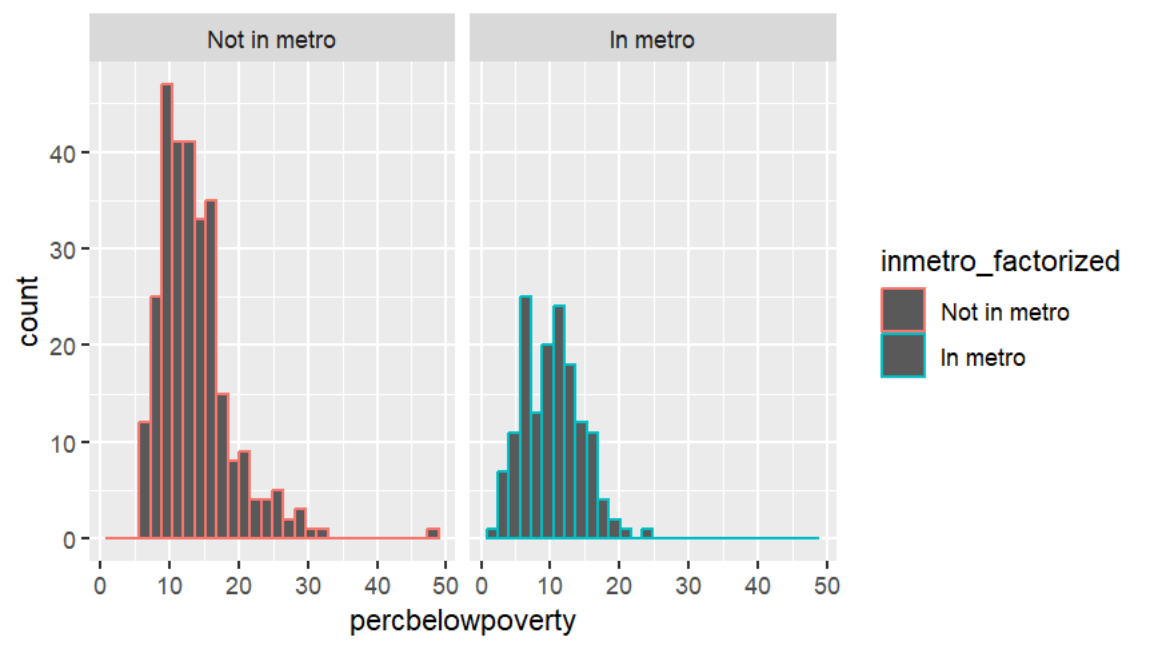
Histogram



Histogram


count
60
40
20
0
0
10
20
30
40
50
percbelowpoverty
state
IL
IN
MI
OH
WI

Histogram


Not in metro
In metro
count
0
10
20
30
40
0
10
20
30
40
50
percbelowpoverty
inmetro_factorized
Not in metro
In metro

Tools

- Pandas
- matplotlib
- Seaborn

Pandas

Pandas is a **Python library** used for **data manipulation, analysis,**

Visualization.

It provides **fast, flexible**, and **easy-to-use** data structures like **Series** and **DataFrame**.

Pandas

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- **Key Features:**

DataFrame: Handles **structured data** (tables, spreadsheets).

Data Cleaning & Transformation: Missing values, filtering, grouping, and merging.

Data Analysis & Statistics: Summary statistics, correlations, and aggregations.

Data Import & Export: Works with **CSV, Excel, SQL, JSON, and more**.

Importing = import pandas as pd

df = pd.read_csv(“path.csv")

Matplotlib

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Matplotlib is a **Python library** used for **creating static, animated,**

Interactive visualizations.

It provides **highly customizable** plotting functions similar to MATLAB.

```
import matplotlib.pyplot as plt
plt.plot()
```

Seaborn

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•**Seaborn** is a **Python library** built on top of **Matplotlib**, used for **statistical data visualization**.

•It provides **informative, and easy-to-read visualizations**

Attractive color palettes & styles.
Regression plots, distribution plots, heatmaps, etc.
Handles DataFrames.

Importing:

```
import seaborn as sns # Import Seaborn
sns.regplot()
```

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Thank you